

LegalForecastBench: Forecasting Judicial Decisions as a Test of Legal Reasoning Ability

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Abstract

Frontier agentic models have recently demonstrated enormous gains in capabilities in a number of domains, but particularly in domains like math and coding where model outputs on even high-level tasks can be verified automatically. Capabilities in the legal domain have advanced as well, but noticeably more slowly than capability levels in verifiable domains, with human lawyers continuing to outperform frontier models on many higher-level legal judgment tasks. The demonstrated importance of verifiable rewards to improving frontier model capabilities raises the question of whether legal judgment tasks can be decomposed into objectively verifiable tasks that could be used to evaluate or train models on higher-level legal reasoning capabilities.

Existing benchmarks for evaluating high-level legal output largely use human-written rubrics that test model outputs. When models were still acquiring basic legal knowledge, simple, bar exam-style questions provided a bounded way to evaluate that knowledge. As capability levels have increased, one approach has been to create new human-written rubrics; Harvey’s Legal Agent Benchmark (LAB), for example, contains 1,200 legal tasks and more than 75,000 rubric criteria that are intended to test what an ideal output would look like. But my preliminary AI-assisted audit of LAB’s litigation tasks (my area of specialty as a practicing lawyer) flagged hundreds of criteria for possible legal errors or misspecification; verification of those findings is ongoing. Moreover, my deep dive into a specific task environment identified concerns about the realism of the synthetic documents and the alignment between the task and its rubric. For example, the supplied documents include programmatic-generation metadata and a stale table-of-contents instruction, and certain criteria prescribe legal conclusions without adequately distinguishing stronger from weaker arguments. These observations raise questions about whether the rewarded conduct will transfer to real-world litigation. They do not establish a benchmark-wide error rate or show that rubric-based training is ineffective.

I argue that judicial decisions in real-world litigation matters supply an external signal that allows evaluation of some high-level legal judgment tasks against an objectively verifiable real-world ground truth. While lawyers and litigants may disagree on what the law is on a certain point, or how it applies to a specific fact pattern, a court’s ruling authoritatively resolves such disputes. Unlike the opinions of expert annotators creating synthetic environments, moreover, judicial decisions have the force of law (and analyzing and understanding such decisions is one of the primary mechanisms by which human lawyers are trained). LegalForecastBench is an example of one way judicial dispositions can be used for more objective evaluation of frontier models’ legal reasoning capabilities. Models receive the available pre-decision litigation records for a motion to dismiss, and are asked to estimate the probability that the presiding judge will dismiss each challenged claim against each moving defendant. The beta includes 91 cases and 387 scored claim–defendant units. In the initial run, GPT-6 Sol has the lowest micro-Brier loss (0.118) and the highest accuracy (84.5%) of seventeen model configurations tested. All seventeen configurations achieved lower micro-Brier loss than a retrospectively fitted

constant-probability reference, but confidence and calibration vary widely, and several strong models make consequential high-confidence errors. We describe how the approach could be extended to model training; this study reports evaluation results, not a completed training experiment.

1 Introduction

Reinforcement learning (RL) techniques have led to significant recent gains in frontier model capabilities, but the most noteworthy gains have been in domains like mathematics and coding that are susceptible to automated verification. Reinforcement learning with verifiable rewards (RLVR) has allowed frontier models to gain capabilities by receiving reward signals from automated correctness checks (like mathematical answer matching, executable tests, or formal proof checking) [1, 2]. Earlier self-play systems demonstrated that automated outcome feedback, in games with explicit rules and automatable scoring mechanisms, allowing models to achieve superhuman performance with sufficient training [3]. Recent announcements from frontier labs have demonstrated the incredible power of these techniques in solving hundreds of longstanding mathematical problems that eluded human solutions.

Frontier models have also made impressive gains in the legal domain, but progress there appears to have been limited to capabilities that are susceptible of some degree of automated verification—meaning that models today cannot reliably perform high-level legal judgment tasks that are difficult to verify. Frontier models have gained excellent legal knowledge. Existing benchmarks that test knowledge of well-settled legal propositions, or their application to simple fact patterns, are easily verifiable (and frontier models likely gain significant legal knowledge in pre-training), so frontier models have become excellent at answering legal questions. And various techniques to ensure that answers to legal research tasks are grounded in sources have largely eliminated earlier problems with hallucinations and made frontier models excellent legal research assistants. But models have increasingly saturated legal benchmarks that focus on bounded tasks with objectively scored answers, like the Multistate Bar Examination [13]. To train models on higher-level legal tasks, researchers have tried at least two approaches: preference evaluation (comparing which of two outputs practicing attorneys prefer) and rubric-based reinforcement learning (grading model outputs against specified criteria to produce a reward). [7]. There are limits to both approaches, as we will discuss in Section 6.4.

This paper aims to demonstrate that judicial decisions in litigation matters provide a useful external signal that can create high-level legal judgment tasks that are objectively verifiable against an external ground truth. This approach rests on a simple fact about the legal system: "It is emphatically the province and duty of the Judicial Department to say what the law is." *Marbury v. Madison*, 5 U.S. 137, 177 (1803). Litigated matters almost always involve disputes about how to interpret, or apply, abstract legal principles to a specific fact pattern. A court's judgment on such a dispute represents the resolution of the matter, and has the force of law. LegalForecastBench attempts to illustrate one specific example of how this external external signal can be utilized to create evaluations that assess the legal reasoning and analytical abilities of frontier models against an objectively verifiable ground truth.

As discussed herein, the alternatives of preference evaluation or rubric-based reinforcement learning are useful techniques for certain kinds of training and evaluation, but both approaches rest on human labels applied by experts whose opinions have no legal authoritative or precedential value. Moreover, as discussed herein, after noticing legal errors in several rubrics, I used frontier models to

audit the litigation-related tasks in Harvey’s Legal Agent Benchmark, currently the most widely cited benchmark for legal, and models identified hundreds of potential errors in rubrics or task specifications. While my work to verify the models’ findings is ongoing, a review of a small sample indicates that most of the flagged issues are indeed genuine legal errors in the rubrics or task specification. These issues illustrate some of the practical challenges of trying to train models on legal tasks using artificially constructed environments.

LegalForecastBench demonstrates that real-world litigation matters can be used to provide models with realistic, challenging high-level judgment tasks that have an objective resolution. Although this paper uses such tasks only for model evaluation, I have developed a separate training data set and am separately exploring whether training on this data can also help improve models’ legal reasoning abilities.

2 Related work

The approach here is consistent with previous research that (i) used older AI techniques to build smaller, custom models for predicting the outcome of litigation (especially U.S. Supreme Court decisions), and (ii) more recent efforts to use frontier models to predict the outcomes of events defined in prediction markets.

2.1 Legal outcome prediction

Using AI models and other techniques to predict litigation outcomes has a long lineage. Ruger and colleagues used hand-coded metadata and statistical techniques to predict the 2002 Term of the U.S. Supreme Court [33]. Later work incorporated more sophisticated machine-learning and natural language processing techniques. Aletras and colleagues in 2016 used a text-based approach to predict the outcomes of European Court of Human Rights decisions [35]. Katz, Bommarito, and Blackman in 2017 used a random-forest classifier over structured Supreme Court data to predict Justice votes and case outcomes. [34]. CAIL2018 used a dataset of more than 2.6 million Chinese criminal cases to predict charges and sentences, with text-classification models providing initial baselines. [36]. As the transformer architecture became dominant, later legal prediction systems increasingly used models like BERT and T5. CLC-UKET, released in 2024, compared fine-tuned BERT and T5 models to zero- and few-shot GPT-3.5 and GPT-5 to predict U.K. Employment Tribunal cases. [38]. The field has thus progressed from statistical models over hand-coded metadata through conventional MLP and supervised machine learning to transformer-based classifiers and, more recently, large language models. Much of this literature differs from the approach studied here, because the models operate on structured metadata or facts and claims extracted from judicial opinions. The approach here simply asks frontier models to reason about the actual pre-decision litigation record, essentially the documents filed in court before the judge who decided the motion, without any access to statistical data or models showing prior decisions.

2.2 Legal AI benchmarks and evaluation

One of the most impressive academic benchmarks is LegalBench, a collaborative benchmark to which 40 authors contributed 162 tasks that aimed to assess legal domain expertise. Although some of the legal issues tested could give rise to disputes in real litigation (for example, there are tests of whether a given statement is hearsay or not), the benchmark deliberately uses relatively clean examples that can be reduced to a binary answer. [12]. The hearsay tasks, for example, explicitly

articulate what proposition a given statement is being used to prove or disprove, and then ask whether that use is hearsay; in real-world litigation, what proposition the statement is being offered to prove may itself be disputed so providing a clean prompt on that point narrows the task to a relatively mechanical application of an explicit rule.

Most recent benchmarks test higher-level judgment tasks that lack an objective ground truth. JudgmentBench, released in May 2026, and Harvey’s LAB targeted open-ended professional legal work, like drafting, analysis, and advice, for which there are potentially multiple defensible outputs and no independently verifiable ground truth. [7, 39]. These benchmarks take different approaches to reward specification. JudgmentBench explicitly compared rubric-based scoring to comparative judgment (pairwise preference judgments collected from practicing attorneys), and found that comparative judgments recover the intended quality ordering substantially better than rubrics. LAB relies on a rubric-based approach. Preference-based evaluation is a form of RLHF that can amplify human biases and errors into the rewards signal [31, 32]; for example, reviewers could reward briefs that are longer, sycophantic, overly confident, or more consistent with the reviewer’s own judicial philosophy or political preferences, rather than rewarding briefs that are most effective at persuading a judge.

The correct scoring rules are those that encourage honest probabilities in expectation [18, 19]. Statistical evaluation also must respect dependence among questions sharing a record [44]. OpenAI’s published work on SWE-bench Verified also emphasizes the importance of avoiding specification error (which is relevant to the findings about rubric-based benchmarks, such as the preliminary audit of LAB). After publishing that benchmark in August 2024, OpenAI noted concerns in February 2026 that the benchmark no longer measures the coding capabilities of frontier models because the tests reject correct solutions and models were increasingly contaminated by having seen the open-source repositories often used for training purposes [45, 46]. These specification problems apply both to rubric-based approaches and to our own outcome labels.

2.3 Verifiable rewards, forecasting, and outcome feedback

Frontier models are trained on several different forms of feedback, as described in prior work. InstructGPT in 2022 provided an influential demonstration of reinforcement learning from human feedback [11]. In 2023, Lightman and colleagues showed that, on mathematical reasoning tasks, process supervision (assigning feedback to individual reasoning steps) outperformed outcome supervision, which gives rewards based on whether the final answer is correct [40].

The efficacy of these techniques in verifiable domains has been exceptional. During reinforcement learning, DeepSeek-R1-Zero’s average pass@1 on AIME 2024 rose from 15.6% to 77.9%; the final DeepSeek-R1 system, which used a broader training pipeline, reached the 96.3rd percentile on Codeforces [1]. AlphaProof coupled reinforcement learning with Lean’s formal proof checker and proved three problems from the 2024 International Mathematical Olympiad, including its hardest problem, solved by only five of 609 human contestants [4]. And OpenAI’s September 2026 announcement that it had solved the Navier–Stokes Millennium Prize Problem reportedly relied on an internal model developed through large-scale reinforcement learning, followed by Lean formalization and verification. [5, 6].

More recent work has used externally resolved real-world events to supply an analogous signal outside domains like mathematics. For example, Autocast, released in 2022, assembled forecasting questions from tournaments and a date-indexed news corpus to allow language models to make predictions based on the information a human would have had at the time. [41]. Halawi and

colleagues in 2024 used a retrieval-augmented forecasting system that could search for information, and generate and aggregate multiple forecasts; the system approached the performance of competitive human forecasters [42]. ForecastBench, introduced in 2024, extended this approach into a dynamic benchmark that continually collects questions about unresolved future events and scores models once the event outcome is known [43].

Outcome-Based Reinforcement Learning to Predict the Future showed in 2025 that resolved binary forecasting questions could serve as reinforcement-learning rewards, training a 14-billion-parameter model on more than 100,000 events using adaptations of GRPO and ReMax [23]. OpenForecaster similarly explored training language models directly on resolved forecasting questions [24]. Some research in China has also used frontier models to predict criminal sentences [25]. LegalForecastBench is an effort to apply a similar approach to litigation outcomes, at least as an evaluation metric for now.

3 Judicial outcomes as a target for legal AI evaluation

3.1 Judicial outcomes as externally verifiable events

Litigation converts a messy legal dispute as to which reasonable minds can differ into a concrete and fixed disposition. Courts prize finality, and final judgments are explicitly written with the goal of achieving a clear and objective disposition of the dispute between the parties that leaves no room for further debate or relitigation of the issues the court has resolved. A final judgment on the merits, with appeals exhausted, has preclusive effect and the force of law, even if a litigant could show that the judgment was based on a legal error [17].

While LegalForecastBench does not directly test models' ability to predict the court's final judgment (instead, testing only their ability to predict the first written disposition of a specified motion), that first-level prediction task is a good initial target because many more cases reach the motion to dismiss stage than the final judgment stage (a litigated final judgment is comparatively rare). And the outcome of the motion to dismiss itself is of critical importance in many disputes and potentially the most economically consequential single stage of litigation. In securities fraud class actions, for example (a specialty that results in billions of dollars in settlements paid in the United States every year), the motion to dismiss effectively determines whether the defendant will ultimately have to make a settlement payment to resolve the case or not. While slightly more than half of the filed cases are dismissed before trial, of those cases that are not, almost all end in settlement. (In the past 18 years of filings, for example, only one securities fraud defendant elected to proceed to trial, Tesla, Inc.) [[cite <https://securities.stanford.edu/litigation.html?filter=CASE>

3.2 Forecasting as a component of legal judgment

Litigation prediction is a key subcomponent of legal work, even when litigation is not at all contemplated or likely. While there are many other tasks involved in giving strategic advice to a client (like understanding the client's objectives), outcome prediction is a key subtask, consistent with the MacCrate skills framework and the New York State Judicial Institute's account of counseling and negotiation [22]. Past research shows that even parties who do not intend to litigate with each other bargain "in the shadow of the law" [21]. When parties are negotiating, their best alternative to a negotiated agreement is often to litigate. As a result, the parties' predictions of how a court will resolve their dispute constrain the bargaining range; neither party is likely to accept an agreement that is less favorable than the result they believe they could obtain in court (after taking into

account the costs and risks of litigation).

A simple risk-neutral model illustrates this [49]. Suppose a plaintiff would receive a fixed award $J > 0$ if successful and zero otherwise. Let q_P and q_D be the plaintiff’s and defendant’s respective estimates of the probability of that success, and let $k_P, k_D \geq 0$ be their remaining litigation costs. Assuming each side bears its own costs, ignoring settlement costs, and restricting attention to monetary payoffs, the plaintiff’s expected litigation payoff is $q_P J - k_P$, while the defendant’s expected trial cost is $q_D J + k_D$. A payment s from defendant to plaintiff is weakly acceptable to both when

$$q_P J - k_P \leq s \leq q_D J + k_D. \quad (1)$$

The interval is nonempty when

$$(q_P - q_D)J \leq k_P + k_D. \quad (2)$$

This is an illustrative reservation-value calculation [50, 51]. Where the parties agree about the probability of success, they can enjoy a surplus from settling the case, equal to the avoided costs. The possibility of settlement disappears when the plaintiff’s estimate of success exceeds the defendant’s estimate of the plaintiff’s success by enough that $(q_P - q_D)J > k_P + k_D$. Even when some interval exists where both parties believe they are better off settling, private information or game theoretic considerations/strategic behavior might prevent agreement [51]. Nonmonetary goals, risk aversion, fee shifting, and uncertainty around the actual damages that would be awarded at trial are additional complications that would require a more complex model. Nevertheless, this simple example illustrates how forecasting litigation outcomes can impact even out-of-court negotiations in a broad swath of commercial disputes or transactions.

Notably, better AI forecasts could yield more surplus to litigants, systematically, if such forecasts help parties’ expectations to converge; simply put, if an AI system becomes sufficiently accurate at predicting litigation outcomes that parties are willing to adjust their expectations based on its output, then the parties are more likely to agree on the likely outcome of a trial. Where $q_P > q_D$, narrowing the gap between their estimates reduces $(q_P - q_D)J$, making the settlement condition easier to satisfy and potentially creating a mutually acceptable payment where none previously existed. If the parties adopt a common forecast, $q_P = q_D$, the settlement interval has width $k_P + k_D$: the full amount of the avoided litigation costs. While modern litigation is much better than many historical options used to resolve disputes (such as feud, self-help, trial by ordeal, etc.), reaching a mutually acceptable agreement is virtually always preferable to the costs, risks, and distraction brought on by protracted, adversarial litigation. Improved forecasting of outcomes can thus help parties more realistically understand the weaknesses and risks of their case, and increase the likelihood of settlement early in the matter.

One major limitation of LegalForecastBench, relative to alternatives, is that it focuses primarily on the models’ analytical ability and does not directly evaluate their ability to prepare legal work product. But litigation prediction is at least a subcomponent of many drafting tasks, and could improve models’ skills.

A basic principle observed in the legal profession is that lawyers must confine their argument to the strongest points, not raise every possible point, and generally should put the strongest arguments first in the brief, moving on to points that are less likely to be dispositive later. In *Jones v. Barnes*, the Supreme Court expressly recognized the value of selecting stronger appellate arguments and the risk that adding weaker ones can dilute them [52]. In order to prepare good work product, then, models need to distinguish the potentially dispositive issues from objections or points that are technically valid but less likely to drive the ultimate outcome.

3.3 What judicial forecasting measures

Prediction is not legal correctness. Judicial outcomes reflect doctrine and record facts, but also procedure, judicial behavior, and institutional context. Judge identity remains visible. A metadata-only comparison, judge masking, and a human forecasting baseline are important future tests. Neither fluent rationales nor a correct forecast demonstrate valid legal reasoning. Training to imitate outcomes can reproduce historical disparities or errors; professional advice also has obligations that prediction scores do not capture.

4 LegalForecastBench: Dataset and evaluation protocol

4.1 Prediction target and information set

Let $c \in \{1, \dots, C\}$ index cases and $u \in \mathcal{U}_c$ index scored units within case c . Each unit identifies a challenged claim or a separately resolvable pleaded subclaim against a moving defendant (or a group of similarly situated defendants, for whom the outcome is predicted together since the legal and factual arguments for dismissal are materially identical as to all of them). We used models to identify appropriate units based on the arguments articulated in each motion; an affirmative defense, an element of a legal cause of action, a requested remedy, or a distinct factual theory was not automatically treated as a separate unit. Rather, units generally tracked claims asserted in the complaint (in some cases with a single claim asserted against multiple defendants being treated as different units, if different defendants advanced different grounds for dismissal). The model receives the pre-decision record that the judge received X_c and the unit specification U_{cu} and returns

$$p_{cu} \in [0, 1], \quad p_{cu} \approx \Pr(Y_{cu} = 1 \mid X_c, U_{cu}), \quad (3)$$

where $Y_{cu} = 1$ means that the judge dismissed that claim-defendant unit in its entirety, in the target motion’s first written disposition. That event is not a final victory in the lawsuit, since a plaintiff might be given an opportunity to file an amended complaint with additional allegations supporting a dismissed claim, other claims might survive even if some are dismissed, and the plaintiff normally has a right to appeal dismissal and seek a de novo adjudication of the issue from the circuit court of appeals.

A full dismissal with leave to amend still has label 1 (i.e., a dismissal is coded the same way, even if it is without prejudice and the plaintiff may seek leave to amend the complaint and try again). A partial dismissal of some claims or theories that leaves the frozen unit alive in a material respect has label 0, meaning the claim-defendant unit was not entirely dismissed. Later amendment, settlement, reconsideration, and appellate reversal do not change the target; the model is specifically asked to predict only the trial judge’s ruling on the specific motion to dismiss. Dispositions that cannot be reliably mapped to a unit were excluded from primary scoring. These choices sacrifice some procedural detail for a stable, objectively interpretable forecast event. Even with these limitations, label construction is a substantive legal task in its own right as the labeling models had to analyze the filed motions to determine what claims were being challenged and ascertain from the judge’s ruling how those were resolved. Two labeling models had to agree on the units, and had the option to escalate ambiguous cases to me for review; in practice, the labeling process was iterative in this preliminary run, as we discovered a number of edge cases that needed to be resolved. The process started with 100 cases but seven were eliminated during the labeling process because the order did not decide the frozen claims, one was eliminated because it was a stipulated dismissal, and one was a plaintiffs’ Rule 12(c) motion for judgment on the pleadings rather than a defendant’s motion to dismiss under Rule 12(b).

First written disposition of the frozen unit	Primary label
Entire claim dismissed, with or without leave to amend	1: fully dismissed
Only one nonseparable theory or remedy dismissed; claim remains	0: not fully dismissed
Claim dismissed against one defendant but survives against another	Separate labels for separately frozen units
Disposition does not reliably resolve the frozen target	Excluded from primary scoring

Table 1: Illustrative label semantics, not additional dataset observations. The documented protocol defines unit boundaries before consulting the outcome.

4.2 Cohort construction and labeling

The analyzed set consists of 91 U.S. federal cases with written decisions dated June 30–August 7, 2026. The cases, as labeled, contained 409 claim-defendant units that could be forecasted. Of those, 387 are scored. Of the scored claim-defendant units, 285 were fully dismissed and 102 were not. The shared-summary preparation inventory contains 482 selected documents. The sample was not random; our pipeline favored cases where some of the necessary documents could be obtained from online databases, although we typically had to purchase a number of documents from the federal PACER system to ensure that the task included all the substantive filings on the motion that were before the judge.

Of the 409 claim-defendant units, 387 are scored; 22 units therefore do not enter the headline metrics. The unscored units involved situations where the plaintiff voluntarily withdrew the frozen claim, the order did not cleanly grant or deny the frozen claim, or (for four units) we determined after freezing the unit that the frozen defendant was the wrong target or the motion did not address the frozen claim.

The case construction and labeling pipeline used models as well as human legal oversight from me. We did take pains to maintain separation between *unitization* and *outcome coding*. In particular, we wished to avoid the possibility that the prediction units could leak information about the judge’s ultimate outcome to the models that were performing prediction. AI models first were asked to identify the claim-defendant units that a motion sought to dismiss, based only on the pre-decision materials (the motion and the docket). A later stage was given access to the claim-defendant units and the judge’s order disposing of the motion, to interpret the disposition for each frozen target. The labeling pipeline we used requires structured outcome agreement and supporting disposition excerpts for automatic acceptance; disagreements between models, ambiguity, and structural defects in the cases were routed to me for lawyer review and a decision. I resolved disputed units and corrected labels as necessary. For purposes of the initial release, this process was iterative and we did not track a blinded human-audit error estimate.

4.3 Information separation and contamination

The models predicting outcomes were not given access to the target decision or the outcome. The construction pipeline also screens docket and filing text to avoid outcome leakage. Models had access to document text tools in a sandbox with no Internet access. Model API calls were made with provider-side web search tools disabled where applicable.

The decisions used in the benchmark were handed down after the reported knowledge or training cutoffs for fifteen of the seventeen tested models. Kimi and Muse did not publish training cutoffs, but based on the model release dates, it appears unlikely that those models were exposed to the

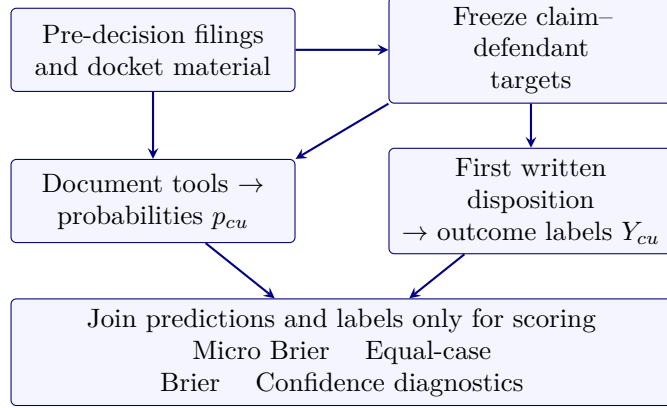


Figure 1: Information separation in the forecasting task. Models inspect pre-decision material and frozen targets. The subsequent judicial disposition is used in the labeling and scoring path, not in the model-visible packet. In the retrospective beta, this is an information boundary rather than a claim that forecasts were made before the decisions occurred.

decisions being tested in training.

The acquisition and labeling pipeline is designed to create new cohorts in the future, when new decisions become available. This enables fresh comparisons even after models might have been trained on the decisions at issue here. However, the downside is that refreshing a benchmark changes its case mix and difficulty; scores from different versions cannot be interpreted as a time series of model improvement.

4.4 Model execution

The main comparison uses a common document-access interface. Models can read documents using tools and produce a probability for each unit, together with a short rationale. Provider-specific reasoning settings are retained in the run registries. Appendix A lists the configuration settings. The study population is the seventeen configurations in Table 2.

4.5 Scoring and statistical inference

For $N = \sum_c |\mathcal{U}_c| = 387$ and $C = 91$, the primary metric is the micro-Brier loss

$$B_{\text{micro}} = \frac{1}{N} \sum_{c=1}^C \sum_{u \in \mathcal{U}_c} (p_{cu} - Y_{cu})^2. \quad (4)$$

The equal-case score changes the estimand by giving each case the same weight:

$$B_{\text{case}} = \frac{1}{C} \sum_{c=1}^C \frac{1}{|\mathcal{U}_c|} \sum_{u \in \mathcal{U}_c} (p_{cu} - Y_{cu})^2. \quad (5)$$

Lower is better for both. Micro-Brier treats each scored claim-defendant unit equally, so cases that involve a motion to dismiss many separate claims receive more weight than cases where only one claim and defendant was subject to a motion to dismiss. Equal-case Brier weights each prediction unit in inverse proportion to the number of units from that case.

The Brier rule is strictly proper for the binary target. If the true conditional event probability is q , then

$$\mathbb{E}[(p - Y)^2 \mid X, U] = (p - q)^2 + q(1 - q), \quad (6)$$

which is minimized by reporting $p = q$. The second term is irreducible uncertainty for that information set. A set of observable judicial events therefore create a verifiable score across the entire probability distribution, even though a single realized outcome standing alone does not reveal whether the individual probability a model assigned to that outcome was correct *ex ante* [18, 19].

We report accuracy at threshold $p \geq 0.5$ as a secondary descriptive metric; in other words, accuracy percentage is simply the percentage of units where the model assigned a 50

The observed dismissal frequency is $\hat{\pi} = 285/387 = 0.73643$. Always predicting dismissal has accuracy 73.64%, so a constant forecast $p = \hat{\pi}$ has micro-Brier $\hat{\pi}(1 - \hat{\pi}) = 0.19410$. The constant is fitted retrospectively on this cohort, so it is a descriptive reference, not a baseline derived from earlier cases. The uninformative forecast $p = 0.5$ has Brier 0.25.

Log loss is a complementary diagnostic:

$$L = -\frac{1}{N} \sum_{c,u} [Y_{cu} \log p_{cu} + (1 - Y_{cu}) \log(1 - p_{cu})]. \quad (7)$$

For a wrong forecast assigning 0.99 to the predicted class, Brier loss is 0.9801 and log loss is approximately 4.605; at confidence 0.90 the corresponding losses are 0.81 and 2.303. Thus log loss imposes a large penalty on extreme errors, specifically, instances where the model assigned a very high probability to the outcome that did not occur.

Claim-defendant units from the same case share facts and legal issues; distinct claims in a single case may suffer from closely related legal defects, so the outcomes are not independent. We therefore measure within-case dependence using an observed-scale intraclass correlation coefficient (ICC) [68].

Across the 387 scored units in 91 cases, the observed-scale outcome ICC was 0.76 (95% CI: 0.61–0.87). Prediction losses also clustered within case because if a model failed to account, for example, for the possibility that a given motion might be denied on procedural grounds, all of the claims in a case that it expected to be dismissed could survive: GPT-6 Sol’s Brier-loss ICC was 0.42 (95% CI: 0.24–0.58). These intervals use 20,000 whole-case bootstrap resamples, and assume independence between cases.

The observed dependence supports resampling whole cases, with model predictions kept paired, when estimating uncertainty in score differences [69]. The seventeen-configuration table is descriptive and does not claim a significance ordering. It does not measure variation across repeated model runs [44].

5 Results

5.1 Frontier models substantially outperform unconditional references

Table 2 reports the seventeen configurations on the ranked leaderboard. GPT-6 Sol achieves the best observed micro-Brier (0.11763), equal-case Brier (0.12949), and accuracy (327/387, 84.50%). All seventeen beat the retrospectively fitted constant micro-Brier of 0.19410; GPT-6 Sol’s micro-Brier is lower than that constant by about 39.4%.

Configuration	Micro Brier	Case Brier	Correct	Acc. (%)	Cost (\$)
GPT-6 Sol	0.11763	0.12949	327/387	84.50	11.01
Claude Fable 5.1	0.12877	0.13653	318/387	82.17	83.18
Grok 4.6	0.13368	0.13480	312/387	80.62	15.72
Claude Opus 5	0.13987	0.13548	310/387	80.10	45.89
GPT-5.6 Sol	0.14009	0.13319	313/387	80.88	24.91
Kimi K3	0.14130	0.15367	303/387	78.29	16.87
GPT-5.6 Luna	0.14322	0.15084	313/387	80.88	1.48
Claude Opus 5.5	0.14540	0.14762	299/387	77.26	20.59
GPT-6.1 Sol	0.14948	0.13885	304/387	78.55	10.68
GPT-6 Astra	0.15221	0.14275	305/387	78.81	59.74
Gemini 3.8 Flash	0.15331	0.13900	310/387	80.10	35.61
Grok 4.7	0.15833	0.16454	298/387	77.00	33.00
Claude Sonnet 5	0.15848	0.15964	293/387	75.71	24.68
GPT-6 Luna	0.16368	0.17379	296/387	76.49	0.62
Claude Sonnet 5.5	0.17298	0.17183	295/387	76.23	6.99
Muse Spark 1.3	0.18121	0.17845	279/387	72.09	8.78
Gemini 3.1 Pro Preview	0.18345	0.18080	302/387	78.04	59.04
Cohort-rate reference	0.19410	—	285/387	73.64	—
Constant $p = 0.5$	0.25000	0.25000	—	—	—

Table 2: Full-document results, ordered by micro-Brier loss. Each model row covers 91 cases and 387 scored units. The last column reports the cost estimate under standard pricing, as detailed in the text. The cohort-rate reference is retrospectively fitted, and its accuracy column uses always-dismiss classification.

Sixteen configurations also beat majority-class accuracy (the 73.64% accuracy of predicting dismissal for every unit). Muse Spark 1.3 does not: 279/387 correct forecasts is 72.09%. Gemini 3.1 Pro Preview has the highest micro-Brier in the panel (0.18345) and still beats that accuracy baseline (302/387, 78.04%).

More expensive or newer configurations do not uniformly score better:

- Astra’s micro-Brier is worse than those of GPT-5.6 Sol and Luna; Astra improves on Luna’s equal-case Brier but not its accuracy.
- Grok 4.7 has worse observed scores than Grok 4.6.
- GPT-6.1 Sol is worse than GPT-6 Sol on micro-Brier (0.14948 versus 0.11763).
- Claude Sonnet 5.5 is worse than Claude Sonnet 5 on micro-Brier (0.17298 versus 0.15848), though its unit accuracy is slightly higher (76.23% versus 75.71%).
- Claude Opus 5.5 is worse than Claude Opus 5 on micro-Brier (0.14540 versus 0.13987) and on accuracy (77.26% versus 80.10%).
- GPT-6 Luna’s full-document scores are worse than GPT-5.6 Luna’s: micro-Brier 0.16368 versus 0.14322, and accuracy 76.49% versus 80.88%.

5.2 Confidence and calibration

Define confidence $h_{cu} = \max(p_{cu}, 1 - p_{cu})$ and the high-confidence subset $H_m = \{(c, u) : h_{cu} \geq 0.9\}$ for configuration m . Comparing the mean confidence in H_m with classification accuracy in H_m gives a useful conditional calibration diagnostic. Of course, this is not a full reliability curve; the

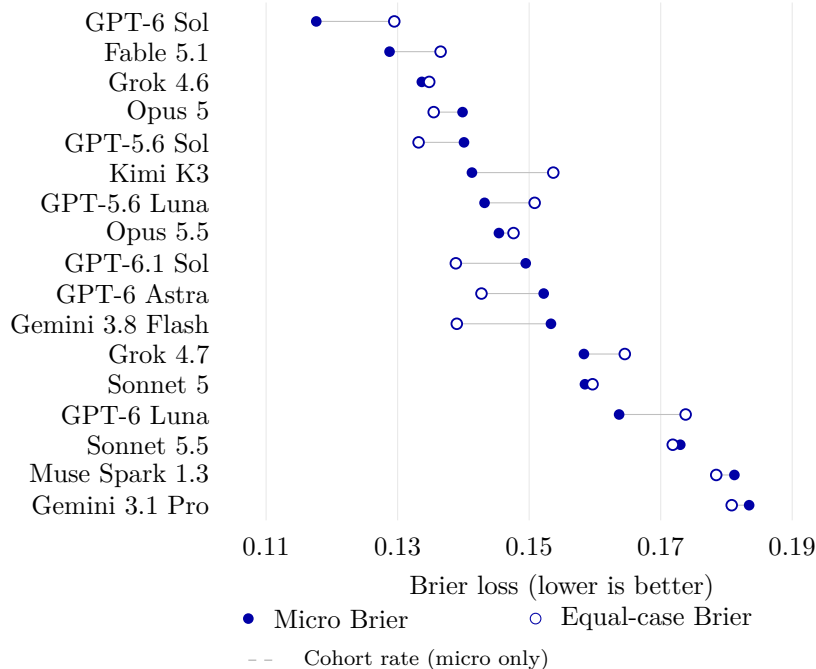


Figure 2: Micro- and equal-case Brier scores for the same seventeen configurations. The connecting segments show weighting sensitivity, not uncertainty intervals. The dashed constant-probability reference applies only to micro-Brier.

subset differs by model, and its units remain clustered by case.

OpenAI models appeared more likely than others to make high-confidence forecasts. GPT-5.6 Sol, GPT-5.6 Luna, and Astra make high-confidence forecasts on 58.9%, 47.0%, and 53.5% of units respectively, compared with 20.9% for Fable 5.1. Their mean confidence in those subsets is approximately 95–97%, while realized accuracy is approximately 89–90%, suggesting some overconfidence, particularly for Astra, which missed 21 of 207 such predictions. Gemini 3.8 Flash also had relatively high sharpness, making 276 high-confidence forecasts (71.3% of units) and missing 35. Gemini 3.1 Pro Preview is sharper still, making 320 high-confidence forecasts and missing 57. GPT-6 Sol provides a useful counterexample within OpenAI, missing 8 of 169 high-confidence forecasts, for 95.3% subset accuracy. GPT-6.1 Sol is sharper and less accurate in that subset, missing 18 of 199. GPT-6 Luna misses 19 of 183. Claude Opus 5.5 misses none of 65.

Other high-performing models were less sharp. For example, Fable has no errors among 81 high-confidence units, while Kimi and Grok 4.6 each miss one. Claude Sonnet 5.5 makes 63 high-confidence forecasts and misses 5. The practical finding is that confidence varies substantially and it materially affects proper scores. Astra’s underperformance relative to cheaper models like Sol, for example, appears largely due to several high-confidence misses, which resulted in large Brier losses.

5.3 Inference cost

The normalized successful-workload costs range from \$0.62 for GPT-6 Luna to \$83.18 for Fable. These are not actual invoice charges. For example, Fable was initially run without caching, resulting in an actual cost of \$189.70. The normalized price was reconstructed after the fact using turn-by-turn token counts, with the reconstruction validated against real billing data from other Anthropic model

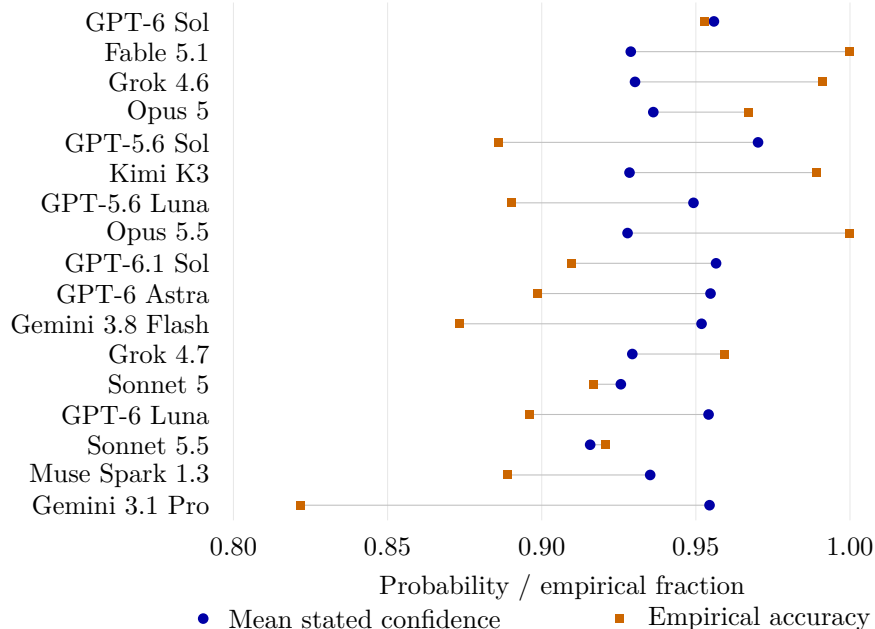


Figure 3: Conditional confidence diagnostic on predictions with $\max(p, 1 - p) \geq 0.9$. Each row uses the subset selected by that model, rather than a common set of units. Wrong/selected counts: GPT-6 Sol: 8/169; Fable 5.1: 0/81; Grok 4.6: 1/112; Opus 5: 4/121; GPT-5.6 Sol: 26/228; Kimi K3: 1/92; GPT-5.6 Luna: 20/182; Opus 5.5: 0/65; GPT-6.1 Sol: 18/199; GPT-6 Astra: 21/207; Gemini 3.8 Flash: 35/276; Grok 4.7: 4/98; Sonnet 5: 5/60; GPT-6 Luna: 19/183; Sonnet 5.5: 5/63; Muse Spark 1.3: 15/135; Gemini 3.1 Pro: 57/320. This is not a full reliability diagram or an estimate of statistical significance.

runs. In several other cases, provider discounts that we received were normalized to standard API rates. On these standardized estimates, GPT-6 Luna, GPT-5.6 Luna, and GPT-6 Sol form the observed micro-Brier cost–score frontier.

5.4 Evaluating Jev, anon-autoregressive discriminative decision model

Jev is described by its developer as a probabilistic “System One” model, exposing native Boolean probabilities rather than generating text or using tools agentially as frontier models do [47]. Because Jev’s context window and lack of agentic tool calling differ from the document-enabled configurations used with frontier models, Jev was evaluated as a separate experiment.

The full set of documents relevant to motions to dismiss is frequently much longer than Jev’s context window, so GPT-5.6 Luna summarized each document with instructions to preserve material facts, arguments, authorities, and procedural posture without commentary or analysis. Luna received only the motion documents that forecasting models received, to prepare summaries. Jev (and two Luna comparator configurations) then received the summaries prepared by the prior Luna session, and the unit questions, in a single request per case, without document tools or multiple turns. The conditions compare Jev through the `typesafe-ai/jev` Gateway alias, GPT-5.6 Luna with high reasoning, and GPT-6 Luna without reasoning.

Jev was not well-calibrated for this task. Jev’s micro-Brier of 0.36754 is worse than the 0.25 loss of always reporting $p = 0.5$. Its mean predicted dismissal probability was approximately 0.305, while the observed frequency in this data set was 0.736. This resulted in severe underprediction in the aggregate. Its 32.6% threshold accuracy is also below majority-class prediction.

Condition	Micro Brier	Case Brier	Acc. (%)
GPT-5.6 Luna full documents	0.14322	0.15084	80.88
GPT-5.6 Luna summary + high reasoning	0.15489	0.16119	78.55
GPT-6 Luna summary + no reasoning	0.17152	0.18694	74.16
Jev summary + one-shot	0.36754	0.34416	32.56

Table 3: Shared-summary results and the full-document Luna reference. All rows use the same 91 cases and 387 scored units. Only the last three rows share a one-request summary interface; the first row is a descriptive full-document reference.

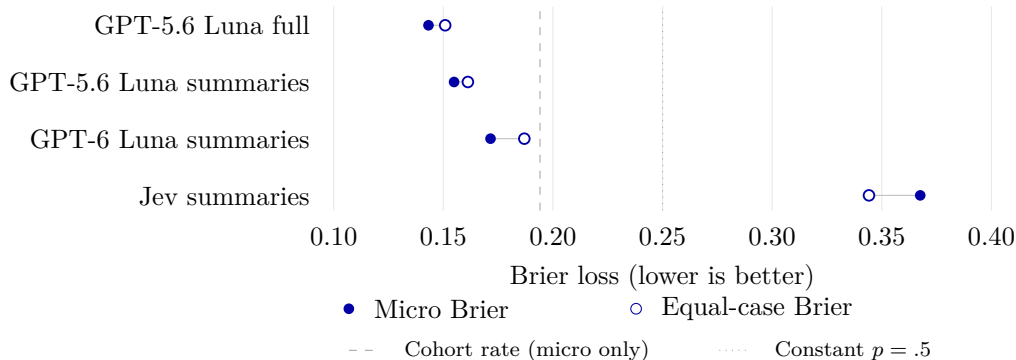


Figure 4: Probability quality with a shared summary cache. Both Luna conditions exploit predictive signal in the summaries, while Jev scores worse than the constant $p = 0.5$ forecast. The full-document reference changes both information access and interaction. Connecting segments indicate weighting sensitivity, not uncertainty intervals.

We tested Luna using the same summaries that Jev received in order to determine how much of Jev’s underperformance was attributable to receiving only summaries of the litigation documents, rather than the full documents that we used in our primary evaluation. However, Luna performed much better than Jev even on this comparison. GPT-5.6 Luna with the identical summaries achieves micro-Brier 0.15489 and accuracy 78.6%. The paired case-cluster bootstrap gives a Luna-minus-Jev difference of -0.21265 , with a 99.44% Bonferroni-adjusted interval (95% familywise coverage) of $[-0.32644, -0.06433]$ across the three model pairs and three metrics. The result remains exploratory because potential cross-case dependence was not assessed.

The shared-summary control shows that the summaries contain useful predictive signal that the Jev configuration does not exploit. Of course, that does not indicate that every important detail survived the summarization process. Luna performed better in the primary evaluation, where it had full document access, than it did in this comparison, with only summaries. That could mean either that the summaries lost some relevant signal, or that the interactive environment (where Luna could choose which documents to read) allowed better performance, or some mix of the two.

Although this experiment indicates that Jev is inferior to frontier models on this particular task, it does not establish that probabilistic or non-generative architectures are unsuitable for litigation prediction or that they will necessarily underperform. It is possible that Jev was not exposed to sufficient training data for this task, or that its architecture or interface was not well-suited to the task, or some combination of the three.

Summary preparation cost an estimated \$10.32. Successful inference estimates are approximately \$0.044 for Jev, \$1.62 for GPT-5.6 Luna (with high reasoning), and \$0.103 for GPT-6 Luna (which

was run without reasoning as a control).

6 Alternative evaluation signals: preferences and rubric-based rewards

6.1 Evaluating open-ended legal work

JudgmentBench compares two ways of evaluating open-ended legal work: asking practicing attorneys which of two outputs they prefer, and asking them to score outputs against specified criteria. Across 30 legal tasks with outputs at three constructed quality levels, comparative judgments recovered the intended quality ordering substantially better than rubric scores and required less than half the annotation time [7]. The constructed quality levels and task selection limit what this establishes about professional work more broadly. Together with judicial outcomes, these approaches provide three distinct evaluation signals: what a court did, which work product an expert prefers, and whether an output satisfies specified requirements.

Preferences and rubrics can assess aspects of advice, analysis, and drafting that a forecast score does not directly measure. Rubrics make the requirements explicit; comparative judgments let reviewers weigh the work as a whole without an exhaustive checklist. Either can supply evaluation data, and either can be used to construct training rewards. Collecting preference judgments for evaluation is not itself reinforcement learning.

Harvey’s Legal Agent Benchmark (LAB) provides a large-scale case study of the rubric approach. Launched on May 6, 2026, it contains 1,200 tasks spanning 24 practice areas and more than 75,000 criteria, with LLM judges scoring model outputs [39, 55]. The LAB-AA partnership with Artificial Analysis extends its use in evaluation [56].¹ Harvey and Applied Compute also report full-parameter reinforcement learning of a legal agent using rubric-based rewards [9]. Harvey’s August 2026 Tenet research preview reports gains on LAB and other legal benchmarks after training a model based on Kimi K3 [10]. These are company-reported training results; they make the quality of the criteria consequential for learning as well as evaluation.

6.2 Preliminary audit of LAB

A rubric can assess richer work than a judicial forecast, but that broader coverage is useful only if the rubric specifies the task correctly and the grader applies it reliably. Errors can arise in the requested work, the supplied record, the legal answer prescribed by a criterion, or the grader’s treatment of a model response. These are distinct failure modes. The underlying question is one of construct validity: whether the score measures the capability the benchmark claims to assess [28].

Our retained AI-assisted review covers the pinned litigation-and-dispute-resolution subset: 52 tasks and 2,858 criteria. GPT-6 Sol flagged 616 criteria; Claude Opus 5.5’s final reconciliation flagged 564. Their union contains 835 flagged criteria, leaving 2,023 unflagged by either final report. Of 345 overlapping flags, both reviewers called 83 problematic and 135 arguable; 127 received different levels of concern. The union represents 29.2% of the 2,858 criteria; the other 70.8% were not flagged, which does not establish that they are correct. These are review outputs, not a verified error

¹Harvey’s product announcements report LAB all-pass results for Opus 4.8 (10.4%, May 28), Fable 5 (13.3%, June 9), and Sonnet 5 (5.8%, June 30) [57–59]. These are Harvey’s evaluations of third-party models, not evidence that each developer independently adopted LAB as its definitive legal benchmark. Prominence does not establish evaluation validity.

rate. Opus’s reconciliation saw Sol’s findings, so agreement in the final reports is not independent confirmation [26].

If a defective criterion supplies a training reward, it also creates a problem of reward misspecification. Reward hacking is a possible learned response; reward-model overoptimization is a possible divergence between an improving proxy score and the underlying target [29, 30]. Our audit examines tasks and criteria, not a training trajectory. It therefore cannot establish that rubric-based RL is ineffective, that synthetic tasks generally harm performance, or that training on litigation tasks caused degradation.

OpenAI’s 2026 analysis of SWE-bench Verified provides a useful methodological analogy: its selected audit of 138 difficult tasks reported problems involving test requirements and task specification [46]. That finding supports examining alignment between tasks and their scoring rules; it provides no defect estimate for LAB. A stronger legal study would compare matched environments built from synthetic and authentic records, independently adjudicate their criteria, and evaluate transfer to held-out professional tasks.

6.3 A motion-to-dismiss case study

The task `analyze-counterparty-motion-to-dismiss` requests an opposition issues memorandum in a fictional software-services dispute involving alleged pre-contractual representations, implementation defects, and jurisdictional, venue, and pleading defenses. The AI reviewers flagged 16 of its 34 criteria. My targeted review uses the task and supplied documents at the pinned upstream revision recorded by the local audit [27]. The analysis below distinguishes direct record observations from legal interpretations and unverified AI flags; it does not treat every flag or solver failure as a grading defect.

Forum selection and litigation strategy. Complaint paragraph 48 sets out a clause selecting state or federal courts in Travis County, Texas, while the action is filed in the Northern District of Georgia. Paragraphs 18–19 assert statutory venue in Georgia, and paragraph 123 argues that the statutory deceptive-practices claim falls outside the clause. Those arguments may be weak, but the complaint is not wholly silent about the forum issue. A rigorous task should ask whether transfer is likely, which claims the clause covers, and whether a procedural objection changes the client’s position. Under *Atlantic Marine*, a valid clause selecting another federal forum is generally enforced through section 1404(a), while a state or foreign forum implicates forum non conveniens [60]. Correcting an opponent’s procedural vehicle is not the same as establishing that the chosen Georgia forum will retain the dispute. A criterion that emphasizes the former without measuring the latter can reward issue identification without strategic assessment.

Accrual and pleaded dates. Criteria C-013–C-015 require an accrual dispute and alternative discovery dates. The complaint alleges discovery of integration problems in August 2022, latency problems by November 2022, an acknowledgment in January 2023, and an unsuccessful patch on April 15, 2023. Paragraph 83 alleges that the plaintiff could not know the full extent of the breach until the patch failed, claims reliance on a promised fix, and states a March 22, 2024 filing date. These allegations require analyzing knowledge sufficient for accrual, distinct breaches, tolling or estoppel, and enforceability of the one-year contract term. Texas law itself restricts contractual limitations periods, subject to statutory exceptions [61].

The appropriate critique is therefore not that the dates mechanically prove the entire claim time-

barred. Nor is it enough to announce that knowledge is always a fact issue immune from dismissal. A plaintiff’s allegations can establish an affirmative defense on the face of the pleading, while a genuine factual or legal issue may prevent that result [62]. The rubric should require applying the governing rule to the actual chronology and contract. It should allow the model to distinguish stronger from weaker opposition arguments instead of requiring the conclusion that every accrual question is unresolved.

Integration versus non-reliance. Criterion C-012 requires the model to characterize the merger-clause defense as legally incorrect. MSA section 14.7 also contains express non-reliance language, which is analytically distinct from an ordinary integration clause. Texas authority distinguishes these contractual concepts and examines the language and surrounding circumstances [63]. The source record therefore does not justify a categorical rubric requiring the ordinary-merger-clause answer without that analysis. This observation identifies an overbroad criterion; it does not establish that the particular disclaimer is enforceable or that the fraud claim must fail.

Hidden output requirements. C-028 requires concrete opposition recommendations for at least eight issues, although the prompt supplies no minimum count. C-022–C-025 prescribe high-severity treatment of specified issues. These criteria permit equivalent severity language; it would be inaccurate to say they necessarily require literal labels. C-026 prescribes organization tied to the motion’s framework and expressly permits omitting Rule 12(b)(1) when no genuine issue exists. Whether those requirements improve usefulness to counsel is a measurement question. They should not be confused with independently established legal errors.

Document construction and limits of the review. The supplied documents have programmatic-generation metadata and include a stale table-of-contents instruction in the agreement. The motion nevertheless contains a caption, table of contents, and table of authorities. We have not established a court-rule violation warranting striking it, proven LLM authorship, or estimated how often such artifacts occur across LAB. Synthetic design can be useful for controlled testing; the concern is whether the rewarded conduct transfers to actual matters. The local audit also records genuine solver mistakes and unresolved legal disagreements. A benchmark-wide conclusion would require a representative expert review of source documents, criteria, model outputs, and grading decisions.

6.4 Comparing supervision signals

Judicial outcomes. Outcomes provide an external reference that the evaluator does not choose. The task designer must still define the event, separate pre-decision information from later evidence, and interpret the disposition correctly. Once that target is fixed, a proper scoring rule evaluates the forecast against what happened without requiring an expert to prescribe the best legal argument. The coverage is narrow: predicting a disposition does not directly evaluate the quality of a brief, the usefulness of advice, or the faithfulness of an explanation. A successful forecast may combine doctrinal analysis with institutional priors rather than identify the legally correct answer.

Rubrics. Rubrics can directly assess several dimensions of professional work, including factual support, legal analysis, completeness, and compliance with a client’s instructions. Their difficulty is specification: the criteria must distinguish indispensable analysis from optional presentation and allow defensible alternative answers. All-pass aggregation magnifies this problem because

one defective criterion can make an otherwise useful work product fail. Some requirements are indispensable, so all-pass scoring is not inherently inappropriate; the question is whether the required items are valid. Nor does learning from rubric rewards necessarily produce only rubric compliance: Gunjal and colleagues report gains on the multiple-choice GPQA-Diamond benchmark relative to direct Likert-based judge rewards, although that finding does not establish transfer to legal work [8].

Preferences. Comparative judgments can capture professional assessments that are difficult to specify in advance, including whether a memorandum prioritizes the issues that matter. They also depend on reviewer expertise, objectives, and consistency. Disagreement may reflect different defensible judgments or measurement noise, and a preferred output need not be the most accurate or useful one. Research outside law shows that human preferences can favor persuasive agreement with a user’s beliefs over factual correctness, and that optimizing a learned reward can amplify such biases [31, 32]. In legal evaluation, a corresponding concern is that a reviewer might prefer confident or congenial advice to a better-supported adverse assessment. This is a risk to investigate, not a finding that JudgmentBench’s attorney judgments exhibit sycophancy. Preference evaluation and preference-based training must be assessed separately.

These signals are complementary. Authentic case records can support an outcome forecast, a rubric checking necessary legal and factual requirements, and a blinded expert comparison of the resulting advice or drafting. Reporting these assessments separately would make disagreements visible: a model might forecast accurately while giving poor advice, or produce polished work that misses the issue most likely to determine the result. The research question is which signals, individually or together, support useful legal judgment beyond the tasks on which they are scored.

7 Future work: from outcome evaluation to outcome feedback

7.1 Outcome feedback as a training signal

LegalForecastGym implements a separate, single-step forecasting environment intended for post-training experiments. It renders a static litigation packet and receives a structured action containing a dismissal probability and cited evidence spans. Training loaders admit only explicitly designated training collections and reject locked test matters. Episode assembly keeps the label out of the observation. The current phase does not implement a multi-turn litigation simulator or train an agent to take procedural actions.

The checked Gym artifacts contain no completed parameter-update experiment or held-out before/after training comparison. They do contain a live sampling comparison on 50 synthetic development episodes, used to assess output format and select an open-weight sampling model. That run is infrastructure evidence, not evidence of legal forecasting improvement. A separately recorded Gym export of 100 matters and 425 scoreable units is also distinct from the 91-case, 387-unit benchmark analyzed here; those populations are not pooled.

The intended learning study asks whether earlier outcome feedback improves predictions on later, disjoint matters and whether any gains extend beyond forecasting. Parameter learning, in-context feedback, and changes to the surrounding agent system are different interventions. Each needs a matched control and a held-out endpoint. Forecasting literature makes such learning plausible, and OpenForecaster reports calibration transfer to other question-answering benchmarks; those findings do not establish improved litigation strategy [23, 24]. Appendix C specifies a concrete prospective design.

7.2 Testing transfer beyond forecasting

A reduction in forecasting loss on later cases would be evidence of improvement on the prediction task. Transfer would require improvement on additional legal tasks assessed independently of that score. The prospective study in Appendix C would therefore compare the trained model with a matched control on issue prioritization, strategic advice, and drafting using later, disjoint matters. The relevant questions are whether the model selects the strongest arguments, relates its advice to the client’s objectives and litigation risks, and produces work supported by the record and applicable law. Explanations should be evaluated separately from the probabilities they accompany.

Practicing lawyers would assess paired outputs with model identity and training condition concealed and presentation order randomized. Both models would receive the same pre-decision record and client instructions; reviewers would initially assess the work without seeing the eventual disposition to limit hindsight effects. Expert comparisons would be supplemented by checks of factual and legal support, with reviewer disagreement reported. Improvement on these independent assessments would provide evidence of transfer; a better forecast score alone would not. LegalForecastGym provides infrastructure for testing this hypothesis, not a completed learning or transfer result.

8 Limitations

Selection into adjudication. The target population is a set of accessible cases with written motion decisions, not all disputes, all filed cases, or all contemplated motions. Parties settle, abandon claims, amend pleadings, or choose not to sue. Cases reaching a published disposition may differ in uncertainty, stakes, representation, and bargaining conditions. Calibration conditional on reaching this dataset does not establish calibration for counterfactual outcomes in the broader dispute population [50]. At the same time, selection does not prove that learned skills cannot transfer: that is an empirical question requiring an independently defined evaluation population.

Judge, court, and party priors. We have not established how much performance comes from close legal analysis rather than judge, court, party, or case-type priors.

Labels and completeness. Unitization can omit, merge, or split legally distinct targets; dispositions can be ambiguous or procedural. Human oversight helps but does not quantify residual error. The absence of a complete blinded label audit and acquisition/exclusion census limits claims about both label accuracy and representativeness. Excluding 22 unscored units preserves an interpretable target but may also change difficulty. These limitations deserve the same scrutiny as disputed rubrics in another benchmark.

Statistical and execution limits. One successful prediction per unit does not characterize run-to-run variance, operational reliability, or expected total cost. The cohort is small and imbalanced. Case clustering does not resolve unverified dependence among related cases. Reported scores compare model–harness–provider configurations, not isolated model weights.

Temporal and transfer limits. A provider-reported cutoff and disabled browsing reduce specific leakage risks without proving contamination absent. New benchmark versions change the case distribution. Generalization to other jurisdictions, procedural stages, settlement decisions, drafting, or strategic advice is not established by this study. Outcome feedback may improve a narrow probability estimate without improving explanations or real client decisions.

9 Conclusion

LegalForecastBench operationalizes an uncertain but consequential legal judgment as a probabilistic forecast of a later judicial event. In the beta, all seventeen document-enabled configurations outperform a cohort-fitted constant in micro-Brier, while confidence behavior and model ordering differ in informative ways. GPT-6 Sol has the strongest observed scores, and the shared-summary comparison exposes a large gap between Jev and Luna that cannot be explained by the summaries containing no signal.

The broader opportunity is to make legal evaluation more empirically grounded. Real outcomes provide an external scoring reference, but useful training signals still require careful task definition, information boundaries, and legal labeling. The next research question is whether feedback on such forecasts improves performance on later cases and transfers to independently assessed legal work. The present evidence motivates that question; it does not answer it.

Reproducibility, access, and disclosure

This manuscript reports the seventeen-configuration comparison and checks its numerical claims against the retained repository artifacts as of September 30, 2026. The accompanying aggregate data and figure script regenerate the plots without court-document bytes. Code and result artifacts are maintained in the [LegalForecastBench repository](#); access to restricted source documents and private adjudication material is a separate audit process. Availability of code is not equivalent to unrestricted source-data replication.

The author designed the benchmark and supplied domain judgments. AI assistants supported this manuscript’s drafting, literature retrieval, and numerical and source checks. The LAB audit discussed above is explicitly an AI-assisted preliminary analysis; the author’s targeted task review is distinguished from a completed independent audit. No model-weight reinforcement-learning experiment is claimed unless explicitly identified as a result of an external cited study.

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A Model configuration and temporal evidence

The table records the maintained local evidence, rather than certifying provider training histories. “K” denotes a reported knowledge cutoff, “T” a reported training cutoff, and “?” no established cutoff. Month-only dates are compared as the first day of the reported month. The reported cutoffs for fifteen configurations precede June 30, 2026. Kimi and Muse remain qualified. A release date is not used to fill an unknown cutoff.

Configuration	Reasoning	Reported cutoff	Relation to window
GPT-6 Sol	High	2026-04-20 (K)	Precedes window
Claude Fable 5.1	Adaptive/default	2026-06 (T)	Precedes window
Grok 4.6	High	2026-02-01 (K)	Precedes window
Claude Opus 5	Adaptive/default	2026-05 (T)	Precedes window
GPT-5.6 Sol	High	2026-02-16 (K)	Precedes window
Kimi K3	High	Unknown (?)	Unknown
GPT-5.6 Luna	High	2026-02-16 (K)	Precedes window
Claude Opus 5.5	High	2026-06 (T)	Precedes window
GPT-6.1 Sol	High	2026-04-30 (K)	Precedes window
GPT-6 Astra	High	2026-04-30 (K)	Precedes window
Gemini 3.8 Flash	High thinking	2026-03 (K)	Precedes window
Grok 4.7	High	2026-05 (K)	Precedes window
Claude Sonnet 5	Adaptive/default	2026-01 (T)	Precedes window
GPT-6 Luna	High	2026-05-18 (K)	Precedes window
Claude Sonnet 5.5	High	2026-06 (T)	Precedes window
Muse Spark 1.3	High	Unknown (?)	Unknown
Gemini 3.1 Pro Preview	High thinking	2025-01 (K)	Precedes window

Table 4: Configuration and cutoff evidence retained by the benchmark. “Precedes window” is a date comparison under the maintained rule, not an independently verified training-history claim.

The common document interface standardizes access, not every inference setting. Claude Fable 5.1, Claude Opus 5, and Claude Sonnet 5 use adaptive thinking at provider defaults. Claude Opus 5.5 and Claude Sonnet 5.5 pin high effort. The other configurations use the high reasoning or thinking setting recorded in their registries. Provider-default sampling and serving updates may still introduce variation. Full-document predictions and the one-request shared-summary conditions are kept separate throughout the analysis.

B Result provenance

The primary cohort identifier is `cycle-1-91-2026-09-08-luna-r5`. Costs in the main table are the totals recorded on the run receipts from provider-reported token usage and the frozen rates.

The checked source revisions are [LegalForecastBench 58371a0e](#), [LegalForecastCorpus b392ddaf](#), and [LegalForecastGym 4cc0f528](#). The LAB source snapshot is [1dd81403](#). These identify inspected versions; they do not certify every execution against the current source tree.

C A prospective test of outcome feedback

A follow-up would freeze a base model and an earlier training cohort, then compare outcome feedback with a matched control on later, disjoint cases. Related cases should remain together across splits. The primary endpoint would be a predeclared difference in proper score; calibration, failure rates, cost, and reasoning validity would be secondary endpoints. Blinded expert assessment of advice or drafting would test transfer.

For a valid forecast, the implemented Brier-shaped reward is

$$R(p, Y) = 0.02 + 0.98[1 - (p - Y)^2], \quad (8)$$

while invalid output receives zero. The positive affine transformation preserves the proper-score optimum among valid forecasts. The implementation also supplies threshold-based and clipped-log alternatives. Evidence spans are logged rather than rewarded. This design separates a directly scored prediction from an assertion that the accompanying reasoning is legally correct.

For a model emitting a rationale z and forecast p , a candidate objective is

$$\max_{\theta} \mathbb{E}_{(X, U, Y) \sim \mathcal{D}_{\text{train}}, (z, p) \sim \pi_{\theta}(\cdot | X, U)} [1 - (p - Y)^2]. \quad (9)$$

This is an outcome-scored prediction objective, not a simulator that allows the agent to alter the litigation trajectory. Optimizing it need not require reinforcement learning: supervised probabilistic prediction, calibration, and policy-gradient approaches are competing methods to test. It also does not directly reward faithful rationales. Leakage-controlled held-out cases and an independent transfer endpoint are therefore necessary to distinguish generalization from memorized case outcomes or improved probability reporting alone.